UNIVERSITY OF CENTRAL PUNJAB

(Faculty of IT)

Computer Communications and Networks (CCN)

Assignment No 1 **Date: 16 Mar 2026**

Total Marks: 100 **Due Date: 30 Mar 2026**

CLO 2 Questions

1. Capture one TCP and one UDP packet using Wireshark, and show the Hex dumps of the captured packets. For each of the TCP & UDP packet, highlight where the following header fields values exist inside the Hex dump, and also write down the values:

   Source & Destination MAC addresses, EtherType value, Source & Destination IP addresses, Protocol value, Source & Destination Port numbers.

2. Draw the flow chart of UDP iterative server interaction with UDP client.

   Write the few lines of code to declare and initialize a Server socket address structure in Client code.

   Also write the few lines of code to read the IP address and port number from a socket address structure, and print them on stdout.

3. Draw the architectural diagrams of the following types of Home Access networks; DSL, Cable & FTTH, showing logical organization of their various components.

4. Consider a TDM link with frame having 24 slots. If each slot carries one byte, and frame rate is 2000 frames per second, determine the TDM line rate and the data rate of one channel. If an application is allocated two channels to transfer data, how much time will be taken to transfer a file of size 1 MB, when the time taken to set up the circuit is 500 msec.

5. Consider a 1 KB packet, which begins at Host A and travels to Host B, over three links, R1 (500 m), R2 (150km) and R3(800 m). The three links are connected by two routers. Let the propagation speed on all three links be 2.8×10^8 m/s. The data rate of link R1 is 1 Mbps, R2 is 3 Mbps and R3 is 4 Mbps. The packet incurs processing delay and queuing delay of 5 msec and 30 msec, respectively, at each router. Compute the following:

   a. Total transmission delay for one packet.
   b. Total propagation delay for one packet.
   c. Total end-to-end delay for one packet.
   d. Also, if 5 such packets are sent back to back, what will be the total delay involved.

Assignment Guidelines:

*To be done in **groups of 3 students; NO individual submission**.*

*To be handwritten on plain **A4 size paper**; all group members to contribute in writing the assignment.*

***Scanned copy** of assignment to be submitted online on **UCP Portal**; only one group member to submit.*

*Assignment to have a separate **title page,** with **Course Name**, **Assign Num**, **Name**, **Reg Num**, **Section**.*

*Assignment will **NOT be accepted after due date.***